

32 bit

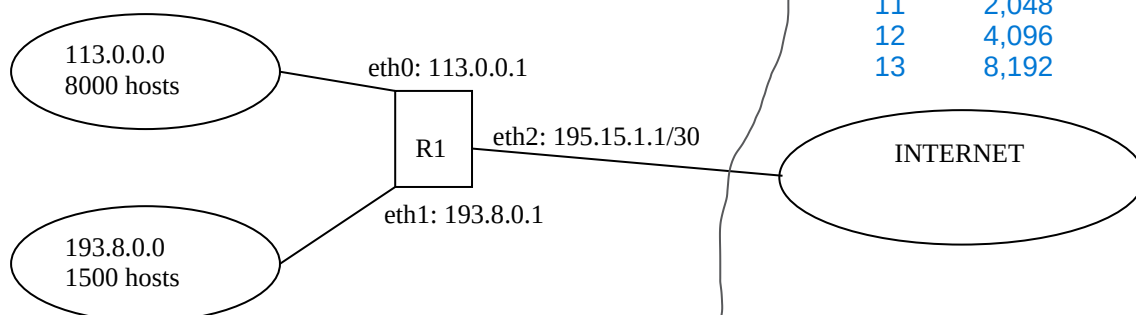
Here he give us number of hosts

Student name, Neptun code: .....

| Power | Value |
|-------|-------|
| 0     | 1     |
| 1     | 2     |
| 2     | 4     |
| 3     | 8     |
| 4     | 16    |
| 5     | 32    |
| 6     | 64    |
| 7     | 128   |
| 8     | 256   |
| 9     | 512   |
| 10    | 1,024 |
| 11    | 2,048 |
| 12    | 4,096 |
| 13    | 8,192 |

## Final Test – sample paper

The commands must be used for Linux Knoppix 6.2!



1./ Write down the commands you would use to configure the network interfaces of the R1 router shown in the figure above.

```
ifconfig eth0 113.0.0.1/19
ifconfig eth1 193.8.0.1/21
ifconfig eth2 195.15.1.1/30
```

NOTE : 8000 need power 13 to be covered  
so we do 32 bit - 13 = 19 to get N of eth0  
1500 needs power 11 to be covered  
so 32.11= 21

2./ After successfully completing the previous step, what routing configuration command can be used to specify that the 500-host network 175.1.0.0 should also be directly accessible on the eth0 interface?

```
route add -net 175.1.0.0/23 eth0
```

3./ Based on the data of the previous two tasks, write down the command you would use to query the routing table of your machine in numerical form, and also write down the data of the displayed table (ie the output of the command)!

```
route -n
```

| Destination | Mask            | Gateway | Interface |
|-------------|-----------------|---------|-----------|
| 113.0.0.0   | 255.255.224.0   | 0.0.0.0 | eth0      |
| 193.8.0.0   | 255.255.248.0   | 0.0.0.0 | eth1      |
| 195.15.1.0  | 255.255.255.252 | 0.0.0.0 | eth2      |
| 175.1.0.0   | 255.255.254.0   | 0.0.0.0 | eth0      |

/19 FOR MASK  
255.255.11100000.0  
.224  
/21 For MASK  
255.255.11111000.0  
.248  
/30 for MASK  
255.255.255.1111100  
.252

do not write  
the /19 ,or  
whatever after /in  
destination

4./ Based on the data from the previous tasks, specify what happens to the following commands.

(Does the packet leave the machine, and if so, on which interface?)

ping 145.7.8.98      the packet will be dropped. unsuccessful ping

ping 175.1.1.1      it will be directed successfully

ping 193.8.71.1      packet will be dropped. unsuccessful ping

NOTE : ping will exit only if its networking address is in networking table

a- we do not have something that start with 145. in destination table

b- 175.1.1.1/23

What is network address :

175.1.0000000/1.1  
.0000000/0.0

Convert 71 from  
decimal to binary  
71 |1  
35 |1  
17 |1  
8 |1  
4 |0  
2 |0  
1 |  
so it is 01000111

c- 193.8.71.1  
 193.8.01000/111.1  
 .01000/000.0  
 193.8.64.0/21

8+8+8+8 32 bit

5./ Our machine is connected to the 155.0.0.0 network. Enter the default NETMASK and BROADCAST values. LAB3 slide 7

class B 155.0.0.0 255.255.0.0 Broadcast address : 155.0.255.255/16

6./ Draw a wiring diagram for a straight-through UTP cable and specify what devices can be used to connect it.

NOTE :  
 TX: the data-sending end, generally plays the role as the transmitter, normally the TXD pin must be connected to the RXD Pin of other devices.  
 RX: the data-receiving end, generally plays the role as the receiver, normally the RXD pin must be connected to the TXD Pin of other devices.



CHECK PHOTO FOR BETTER DRAWING

put circle in PC : 1 and 2  
 put rectangle in PC: 3 and 6  
 put circle in SW : 3 and 6  
 put rectangle in SW: 1 and 2  
 circle is TX  
 rectangle is RX

7./ We have two computers in a common broadcast domain with their IP address:

„PC A”: 196.1.40.3/20,

„PC B”: 196.1.38.9/20.

The machine interfaces work with the specified IP addresses.

Can the address of machine “B” be included in the ARP table of machine “A”? Justify your answer!

196.1.40.3/20  
 196.1.0010/1000.3  
 NA: 196.1.0010/000.0  
 NA: 196.1.32.0/20

196.1.38.9/20  
 192.1.0010/0110.9  
 NA: 192.1.0010/0110.9  
 NA: 192.1.32.0/20

\*NA = network address



JUSTIFY: the 2nd computer mac address can be stored in first computer ARP table only if they had before a successful layer 3 communication or they are in the same network. Because based on our calculations, they are in same subnet of network otherwise it can not be stored

Convert 40 from decimal to binary

40 |  
 20 | 0  
 10 | 0  
 5 | 0  
 2 | 1  
 1 | 0  
 so it is 00101000

8./ Name the following IPv6 addresses / address ranges / prefixes.

::/128 unspecified ip address  
 ::1/128 IPv6 loop back interface address  
 FF00::/8 multicast address  
 FE80::/10 link local address space  
 FF02::1 special multicast group  
 FF02::2 first one is AL host multicast 2nd is AL router multticast  
 ::FFFF/96 IPv4 embedded ipv6 address  
 2001:ABCD::/32 global unicast address

Convert 38 from decimal to binary

38 |  
 19 | 0  
 9 | 1  
 4 | 1  
 2 | 0  
 1 | 0  
 so it is 00100110